

Spatial Analysis of Child Malnutrition in Rwanda

District-Level Integration of DHS Data and Agro-Climatic Indicators

Richard USENGIMANA

2026-05-03

Load Required Libraries

```
library(haven)      # read .dta (Stata) DHS files
library(dplyr)      # data manipulation
library(tidyr)      # reshaping
library(readr)      # read CSV files

library(sf)         # shapefiles and spatial operations
library(terra)      # raster operations (NDVI)

library(spdep)      # spatial weights, Moran's I, LISA
library(spatialreg) # spatial lag and spatial error models

library(ggplot2)    # plots
library(viridis)    # color scales

library(naniar)     # missing data visualization
```

STEP 1: Load RDHS Data

The DHS files are loaded from their standard folder structure. Four files are used: - **KR file**: Child-level anthropometry (HAZ, WAZ, WHZ, child characteristics) - **HR file**: Household-level variables (wealth index, water, sanitation) - **IR file**: Individual/Maternal variables (education, BMI, ANC visits) - **GPS file**: Cluster coordinates and altitude

```
# ADJUST PATHS TO YOUR OWN MACHINE
kr <-
read_dt
a("~/Downloads/Thesis_Material/RW_2019-20_DHS/RWKR81DT/RWKR81FL.DTA")
hr <-
read_dt
a("~/Downloads/Thesis_Material/RW_2019-20_DHS/RWHR81DT/RWHR81FL.DTA")
```

```

ir <-
read_dt
a("~/Downloads/Thesis_Material/RW_2019-20_DHS/RWIR81DT/RWIR81FL.DTA")
gps <-
st_rea
d("~/Downloads/Thesis_Material/RW_2019-20_DHS/RWGE81FL/RWGE81FL.shp")

## Reading layer `RWGE81FL' from data source
##
`/home/guest/Downloads/Thesis_Material/RW_2019-20_DHS/RWGE81FL/RWGE81FL.shp'
## using driver `ESRI Shapefile'
## Simple feature collection with 500 features and 20 fields
## Geometry type: POINT
## Dimension: XY
## Bounding box: xmin: 28.89335 ymin: -2.8127 xmax: 30.82515 ymax: -1.078855
## Geodetic CRS: WGS 84

cat("KR rows:", nrow(kr), "\n")

## KR rows: 8092

cat("HR rows:", nrow(hr), "\n")

## HR rows: 12949

cat("IR rows:", nrow(ir), "\n")

## IR rows: 14634

cat("GPS rows:", nrow(gps), "\n")

## GPS rows: 500

```

STEP 2: Load Rwanda District Shapefile

The district shapefile is loaded from the National Institute of Statistics Rwanda (NISR). It provides the 30 administrative district boundaries used for spatial aggregation.

```

library(geodata)
library(sf)

```

```
# Download Rwanda district boundaries (level = 2 = districts)
```

```

rwanda <- geodata::gadm(country = "RWA", level = 2, path = tempdir())

# Convert to sf
rwanda_sf <- st_as_sf(rwanda)
nrow(rwanda_sf)

## [1] 30

names(rwanda_sf) # confirm which column holds district names (e.g.,
NAME_2)

## [1] "GID_2"      "GID_0"      "COUNTRY"    "GID_1"      "NAME_1"
"NL_NAME_1"
## [7] "NAME_2"      "VARNAME_2"  "NL_NAME_2"  "TYPE_2"      "ENGTYPE_2"
"CC_2"
## [13] "HASC_2"      "geometry"

```

STEP 3: Inspect Missing Values Before Cleaning

Before removing any records we first quantify how many children have flagged or implausible z-score codes (9996–9999 in DHS encoding).

```

# DHS special codes for missing / implausible anthropometric values
dhs_missing_codes <- c(9996, 9997, 9998, 9999)

kr_temp <- kr %>%
  mutate(
    haz_raw = ifelse(hw70 %in% dhs_missing_codes, NA, hw70),
    waz_raw = ifelse(hw71 %in% dhs_missing_codes, NA, hw71),
    whz_raw = ifelse(hw72 %in% dhs_missing_codes, NA, hw72)
  )

cat("Missing HAZ (stunting) before filter:",
sum(is.na(kr_temp$haz_raw)), "\n")

## Missing HAZ (stunting) before filter: 4283

cat("Missing WAZ (underweight) before filter:",
sum(is.na(kr_temp$waz_raw)), "\n")

## Missing WAZ (underweight) before filter: 4281

cat("Missing WHZ (wasting) before filter:",
sum(is.na(kr_temp$whz_raw)), "\n")

## Missing WHZ (wasting) before filter: 4283

```

STEP 4: Clean Z-scores and Create Malnutrition Outcome Variables

Important correction from original code: DHS stores z-scores multiplied by 100 (e.g., -2.0 SD = -200 , -1.5 SD = -150). Therefore the threshold for malnutrition is `z - score < -200` (equivalent to < -2 SD). The original code already applied this correctly. Confirmed here.

```
kr_clean <- kr %>%
  select(caseid, v001, v002, v003, b4, b11, bord, hw70, hw71, hw72,
b5, hw1) %>%
  rename(
    cluster      = v001,
    hh_num       = v002,
    mother_line  = v003,
    child_sex    = b4,
    birth_int    = b11,
    birth_ord    = bord,
    haz          = hw70,
    waz          = hw71,
    whz          = hw72,
    child_alive  = b5,
    child_age    = hw1
  ) %>%
  mutate(
    # Recode DHS missing/flagged codes to NA (codes 9996-9999)
    haz = ifelse(haz %in% c(9996, 9997, 9998, 9999), NA, haz),
    waz = ifelse(waz %in% c(9996, 9997, 9998, 9999), NA, waz),
    whz = ifelse(whz %in% c(9996, 9997, 9998, 9999), NA, whz)
  ) %>%
  # Complete-case analysis: keep only children with all three valid z-
scores
  filter(!is.na(haz), !is.na(waz), !is.na(whz)) %>%
  mutate(
    # Z-scores are stored ×100 in DHS; threshold is -200 (= -2.0 SD)
    stunted      = ifelse(haz < -200, 1, 0),
    underweight  = ifelse(waz < -200, 1, 0),
    wasted       = ifelse(whz < -200, 1, 0)
  )

cat("Clean KR rows after removing invalid z-scores:", nrow(kr_clean),
"\n")

## Clean KR rows after removing invalid z-scores: 3806
```

After filtering, there should be zero NAs in the outcome variables.

```

missing_summary <- data.frame(
  Variable      = c("Stunting", "Wasting", "Underweight"),
  Missing_Count = c(sum(is.na(kr_clean$stunted)),
                    sum(is.na(kr_clean$wasted)),
                    sum(is.na(kr_clean$underweight))),
  Missing_Percent = c(mean(is.na(kr_clean$stunted)) * 100,
                     mean(is.na(kr_clean$wasted)) * 100,
                     mean(is.na(kr_clean$underweight)) * 100)
)
print(missing_summary)

##      Variable Missing_Count Missing_Percent
## 1   Stunting             0              0
## 2    Wasting             0              0
## 3 Underweight             0              0

```

STEP 5: Merge Household (HR) Data

Household wealth index, water source, and toilet type are merged to each child using the cluster + household number key.

```

hr_clean <- hr %>%
  select(hv001, hv002, hv270, hv201, hv205) %>%
  rename(
    cluster = hv001,
    hh_num  = hv002,
    wealth_idx = hv270,
    water_src = hv201,
    toilet    = hv205
  )

kr_hr <- kr_clean %>%
  left_join(hr_clean, by = c("cluster", "hh_num"))

cat("KR clean rows:", nrow(kr_clean), "\n")

## KR clean rows: 3806

cat("After HR merge:", nrow(kr_hr), "\n")

## After HR merge: 3806

```

STEP 6: Merge Maternal (IR) Data

Maternal education, BMI, and ANC visits are merged using the cluster + household number + mother line number key.

```
ir_clean <- ir %>%
  select(v001, v002, v003, v106, v445, m14_1) %>%
  rename(
    cluster      = v001,
    hh_num       = v002,
    mother_line  = v003,
    mat_edu      = v106,
    mat_bmi      = v445,
    anc_visits   = m14_1
  ) %>%
  mutate(
    # DHS stores BMI ×100; convert to actual BMI
    mat_bmi      = mat_bmi / 100,
    # DHS missing codes for ANC: 98 = "don't know", 99 = missing
    anc_visits   = ifelse(anc_visits %in% c(98, 99), NA, anc_visits)
  )

kr_hr_ir <- kr_hr %>%
  left_join(ir_clean, by = c("cluster", "hh_num", "mother_line"))

cat("After HR merge rows:", nrow(kr_hr), "\n")

## After HR merge rows: 3806

cat("After IR merge rows:", nrow(kr_hr_ir), "\n")

## After IR merge rows: 3806
```

STEP 7: Merge GPS Cluster Data (Coordinates + Altitude)

GPS coordinates and altitude are merged at cluster level. Clusters with coordinates (0, 0) are excluded as they represent missing location data in DHS.

```
gps_clean <- gps %>%
  st_drop_geometry() %>%          # drop geometry; we only need the
  attribute table                  # attribute table
  select(DHSCLUST, LATNUM, LONGNUM, ALT_DEM) %>%
  rename(
    cluster = DHSCLUST,
    latitude = LATNUM,
```

```

    longitude = LONGNUM,
    altitude = ALT_DEM
  ) %>%
  # Remove clusters with missing/invalid coordinates
  filter(latitude != 0, longitude != 0)

kr_full <- kr_hr_ir %>%
  left_join(gps_clean, by = "cluster")

cat("After IR merge rows:", nrow(kr_hr_ir), "\n")
## After IR merge rows: 3806

cat("After GPS merge rows:", nrow(kr_full), "\n")
## After GPS merge rows: 3806

# Verify coordinates look sensible for Rwanda
cat("\nLatitude range:", range(kr_full$latitude, na.rm = TRUE), "\n")
##
## Latitude range: -2.8127 -1.078855

cat("Longitude range:", range(kr_full$longitude, na.rm = TRUE), "\n")
## Longitude range: 28.89335 30.82515

```

STEP 8: Final RDHS Dataset (Filter Complete Cases)

We retain only children whose mothers have valid BMI and ANC visit records, as these are key covariates in the spatial regression models.

```

RDHS_final <- kr_full %>%
  filter(!is.na(mat_bmi), !is.na(anc_visits))

cat("Rows after filtering mat_bmi and anc_visits:", nrow(RDHS_final),
    "\n")
## Rows after filtering mat_bmi and anc_visits: 3804

# Save cleaned dataset for reproducibility
RDHS_final_cleaned <- RDHS_final %>%
  select(-any_of(c("geometry", "row.names"))) # remove any leftover
  spatial cols

write.csv(RDHS_final_cleaned,

```

```

~/Downloads/RDHS_final_clean.csv",
row.names = FALSE)

cat("RDHS final dataset saved.\n")

## RDHS final dataset saved.

```

STEP 9: Assign Children to Districts (Spatial Join)

Children are converted to spatial points and joined to the district shapefile so that each child record gains the district name (NAME_2).

```

data <- read.csv("~/Downloads/RDHS_final_clean.csv")

# Create spatial object
children_sf <- st_as_sf(
  data,
  coords = c("longitude", "latitude"),
  crs     = 4326
)

# Reproject children to match district CRS
children_sf <- st_transform(children_sf, st_crs(rwanda_sf))

# Spatial join: attach district name to each child
children_with_district <- st_join(children_sf, rwanda_sf[,
c("NAME_2")])

# Check: how many children have no district match?
cat("Children without district match:",
sum(is.na(children_with_district$NAME_2)), "\n")

## Children without district match: 21

```

STEP 10: Compute District-Level Malnutrition Prevalence Rates

District-level prevalence rates (%) for stunting, wasting, and underweight was computed by averaging the binary outcome variables within each district.

```

district_summary <- children_with_district %>%
  st_drop_geometry() %>%
  group_by(NAME_2) %>%
  summarise(
    stunting_rate = mean(stunted, na.rm = TRUE) * 100,

```



```

wasting_rate      = mean(wasted,      na.rm = TRUE) * 100,
underweight_rate  = mean(underweight, na.rm = TRUE) * 100,
n_children        = n(),
.groups           = "drop"
)

cat("Districts with computed prevalence:", nrow(district_summary), "\n")

## Districts with computed prevalence: 31

Join prevalence rates back to the shapefile for mapping:

district_map <- rwanda_sf %>%
  left_join(district_summary, by = "NAME_2")

cat("Districts in map:", nrow(district_map), "\n")

## Districts in map: 30

cat("Districts with stunting data:", sum(!
is.na(district_map$stunting_rate)), "\n")

## Districts with stunting data: 30

```

STEP 11: Map District-Level Malnutrition Prevalence

```

plot_malnutrition_map <- function(data, variable, legend_title,
map_title) {
  centroids <- st_point_on_surface(data)

  ggplot(data) +
    geom_sf(aes(fill = .data[[variable]]), color = "grey30", linewidth

```

```

= 0.25) +
  geom_sf_text(
    data = centroids,
    aes(label = paste0(NAME_2, "\n", round(.data[[variable]]), 1),
"%"),
    size = 1.8
  ) +
  scale_fill_viridis_c(
    name      = legend_title,
    option    = "viridis",
    na.value  = "white"
  ) +
  labs(
    title     = map_title,
    caption  = "Source: Rwanda DHS 2019/20"
  ) +
  theme_minimal()
}

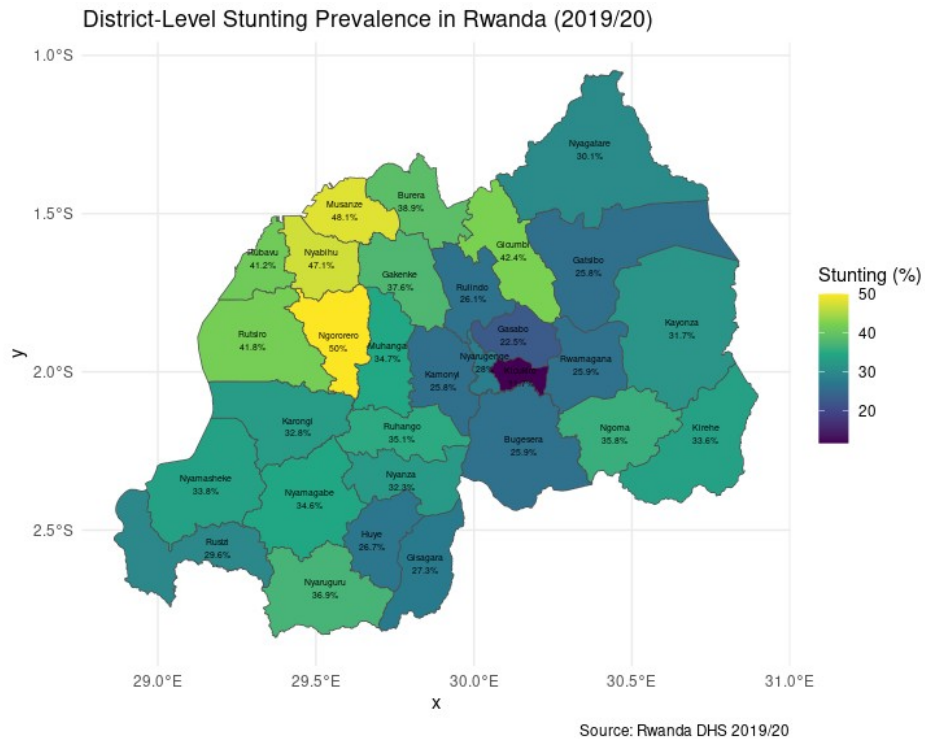
```

Stunting Map

```

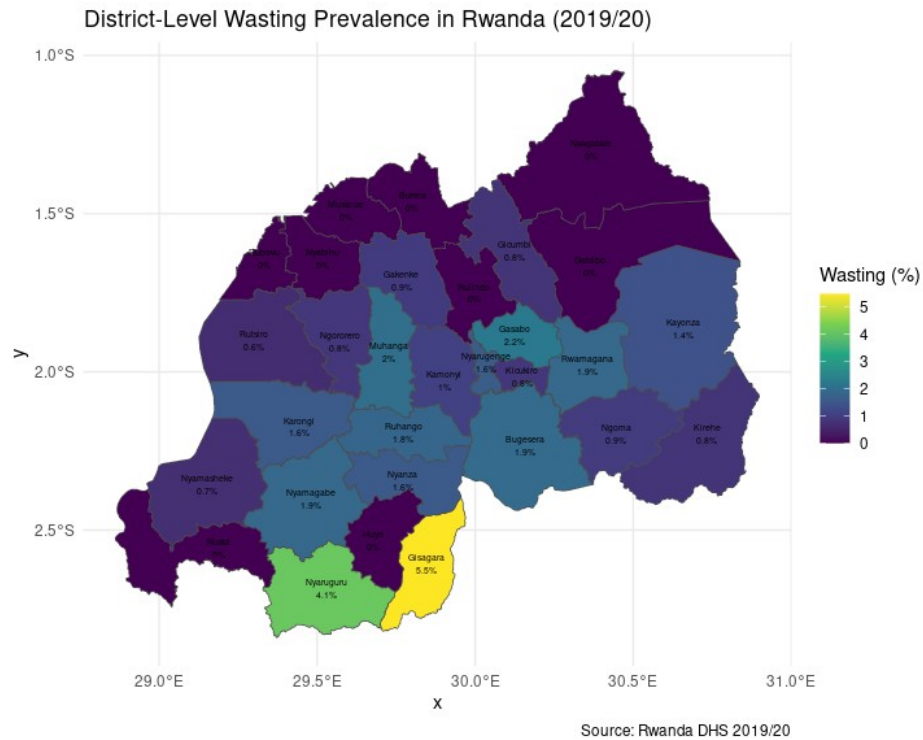
plot_malnutrition_map(
  data      = district_map,
  variable  = "stunting_rate",
  legend_title = "Stunting (%)",
  map_title  = "District-Level Stunting Prevalence in Rwanda
(2019/20)"
)

```



Wasting Map

```
plot_malnutrition_map(
  data      = district_map,
  variable  = "wasting_rate",
  legend_title = "Wasting (%)",
  map_title  = "District-Level Wasting Prevalence in Rwanda
(2019/20)"
)
```



Underweight Map

```
plot_malnutrition_map(
  data      = district_map,
  variable  = "underweight_rate",
  legend_title = "Underweight (%)",
  map_title  = "District-Level Underweight Prevalence in Rwanda
(2019/20)"
)
```



```

) %>%
print()

## # A tibble: 1 × 4
##   total_rows missing_precip valid_precip missing_pct
##       <int>         <int>         <int>         <dbl>
## 1      294000         112800         181200         38.4

# Remove rows with missing precipitation
rainfall_clean <- rainfall %>%
  filter(!is.na(precip))

```

Convert Rainfall Grid to Spatial Points and Join to Districts

```

rainfall_sf <- st_as_sf(
  rainfall_clean,
  coords = c("longitude", "latitude"),
  crs     = 4326
)

rainfall_sf <- st_transform(rainfall_sf, st_crs(rwanda_sf))

# Spatial join: attach district name to each rainfall grid point
rainfall_with_district <- st_join(rainfall_sf, rwanda_sf[,
c("NAME_2")])

# Check coverage
cat("Rainfall points with district match:", sum(!
is.na(rainfall_with_district$NAME_2)), "\n")

## Rainfall points with district match: 168000

cat("Rainfall points without district match:",
sum(is.na(rainfall_with_district$NAME_2)), "\n")

```

```
## Rainfall points without district match: 13200

rainfall_with_district_clean <- rainfall_with_district %>%
  filter(!is.na(NAME_2))

cat("Districts covered by rainfall data:",
    length(unique(rainfall_with_district_clean$NAME_2)), "\n")

## Districts covered by rainfall data: 30
```

Aggregate to District-Level Rainfall Statistics

```
rainfall_district <- rainfall_with_district_clean %>%
  st_drop_geometry() %>%
  group_by(NAME_2) %>%
  summarise(
    rainfall_mean = mean(precip, na.rm = TRUE),
    rainfall_sd   = sd(  precip,   na.rm = TRUE),
    n_rainfall_obs = n(),
    .groups       = "drop"
  )

cat("Districts with rainfall data:", nrow(rainfall_district), "\n")

## Districts with rainfall data: 30

print(rainfall_district)

## # A tibble: 30 × 4
##   NAME_2    rainfall_mean rainfall_sd n_rainfall_obs
##   <chr>          <dbl>         <dbl>         <int>
## 1 Bugesera      80.7           56.7           9600
## 2 Burera       100.           60.2           3840
## 3 Gakenke      97.8           59.9           5280
## 4 Gasabo       88.2           61.5           3360
## 5 Gatsibo      84.3           57.0          11040
## 6 Gicumbi      95.7           62.6           6480
## 7 Gisagara    103.           65.5           1200
## 8 Huye         101.           63.4           4080
## 9 Kamonyi      87.3           58.4           5280
```

```
## 10 Karongi          100.          54.2          6480
## # i 20 more rows
```

Map District-Level Rainfall

```
district_map <- district_map %>%
  left_join(rainfall_district[, c("NAME_2", "rainfall_mean")], by =
"NAME_2")

summary(district_map$rainfall_mean)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  80.00   86.90   96.01   96.76  101.59  126.14

cat("Districts missing rainfall:",
sum(is.na(district_map$rainfall_mean)), "\n")

## Districts missing rainfall: 0

centroids_rain <- st_point_on_surface(district_map)

ggplot(district_map) +
  geom_sf(aes(fill = rainfall_mean), color = "grey30", linewidth =
0.25) +
  geom_sf_text(data = centroids_rain, aes(label = NAME_2), size = 2.2)
+
  scale_fill_viridis_c(option = "viridis", name = "Rainfall (mm)") +
  labs(
    title   = "District-Level Mean Monthly Rainfall in Rwanda (2001-
2020)",
    caption = "Source: CHIRPS 2001-2020"
  ) +
  theme_minimal()
```



```
## NDVI stack layers: 9

print(ndvi_stack)

## class      : SpatRaster
## dimensions  : 208, 221, 9  (nrow, ncol, nlyr)
## resolution  : 926.6254, 926.6254  (x, y)
## extent      : 3225583, 3430367, -305786.4, -113048.3  (xmin, xmax,
ymin, ymax)
## coord. ref. : +proj=sinu +lon_0=0 +x_0=0 +y_0=0 +R=6371007.181
+units=m +no_defs
## source(s)   : memory
## names        : MOD13~d0001, MOD13~d0001, MOD13~d0001, MOD13~d0001,
MOD13~d0001, MOD13~d0001, ...
## min values   :      -2e-05,  -1.998e-05,      -2e-05,      -2e-05,
-2e-05,      -2e-05, ...
## max values   :      1e-04,   1.000e-04,      1e-04,      1e-04,
1e-04,      1e-04, ...
```

Crop and Mask NDVI to Rwanda, then Extract District Means

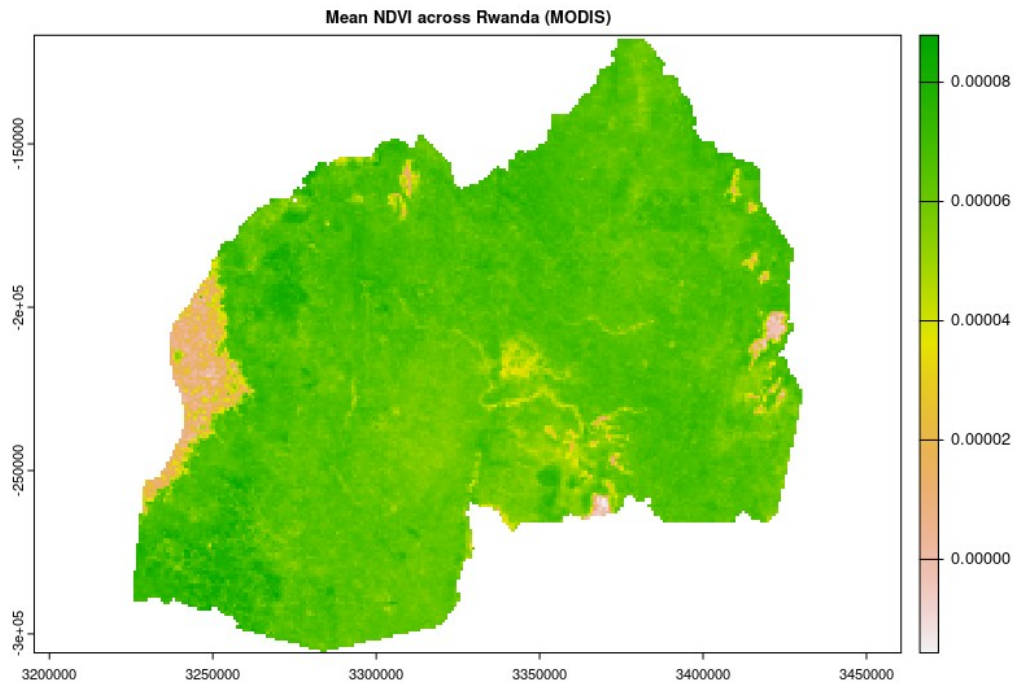
```
# Convert district sf to terra SpatVector and match CRS
rwanda_vect      <- vect(rwanda_sf)
rwanda_vect_ndvi <- project(rwanda_vect, crs(ndvi_stack))

# Crop and mask to Rwanda
ndvi_crop <- crop(ndvi_stack,  rwanda_vect_ndvi)
ndvi_mask <- mask(ndvi_crop,  rwanda_vect_ndvi)

# Compute pixel-wise temporal mean across all months
```

```
ndvi_mean <- mean(ndvi_mask, na.rm = TRUE)
```

```
plot(ndvi_mean, main = "Mean NDVI across Rwanda (MODIS)")
```



```
# Extract district-level NDVI means
```

```
ndvi_district <- terra::extract(  
  ndvi_mean,  
  rwanda_vect_ndvi,  
  fun      = mean,  
  na.rm    = TRUE  
)
```

```
# Attach district names
```

```
ndvi_district$NAME_2 <- rwanda_sf$NAME_2
```

```
# Rename the mean NDVI column
```

```
ndvi_district <- ndvi_district %>%  
  rename(ndvi_mean = 2) %>%  
  select(NAME_2, ndvi_mean)
```

```
print(ndvi_district)
```

```
##      NAME_2      ndvi_mean  
## 1      Burera 6.487165e-05
```

## 2	Gakenke	6.783648e-05
## 3	Gicumbi	6.918227e-05
## 4	Musanze	6.641796e-05
## 5	Rulindo	6.675159e-05
## 6	Gisagara	6.243105e-05
## 7	Huye	6.395313e-05
## 8	Kamonyi	6.082798e-05
## 9	Muhanga	6.435901e-05
## 10	Nyamagabe	6.624524e-05
## 11	Nyanza	6.235965e-05
## 12	Nyaruguru	6.705366e-05
## 13	Ruhango	6.023843e-05
## 14	Bugesera	5.714040e-05
## 15	Gatsibo	6.555128e-05
## 16	Kayanza	6.275848e-05
## 17	Kirehe	6.363870e-05
## 18	Ngoma	6.380351e-05
## 19	Nyagatare	6.615632e-05
## 20	Rwamagana	6.580298e-05
## 21	Karongi	5.943859e-05
## 22	Ngororero	6.678265e-05
## 23	Nyabihu	7.004100e-05
## 24	Nyamasheke	6.441147e-05
## 25	Rubavu	6.245904e-05
## 26	Rusizi	7.519074e-05
## 27	Rutsiro	4.780395e-05
## 28	Gasabo	6.260337e-05
## 29	Kicukiro	5.376454e-05
## 30	Nyarugenge	5.784076e-05

Join NDVI to District Map

```
# Remove any prior ndvi_mean column to avoid duplicates on re-run
district_map <- district_map %>%
  select(-any_of(c("ndvi_mean", "ndvi_mean.x", "ndvi_mean.y"))) %>%
  left_join(ndvi_district[, c("NAME_2", "ndvi_mean")], by = "NAME_2")

cat("Districts missing NDVI:", sum(is.na(district_map$ndvi_mean)), "\n")

## Districts missing NDVI: 0

summary(district_map$ndvi_mean)

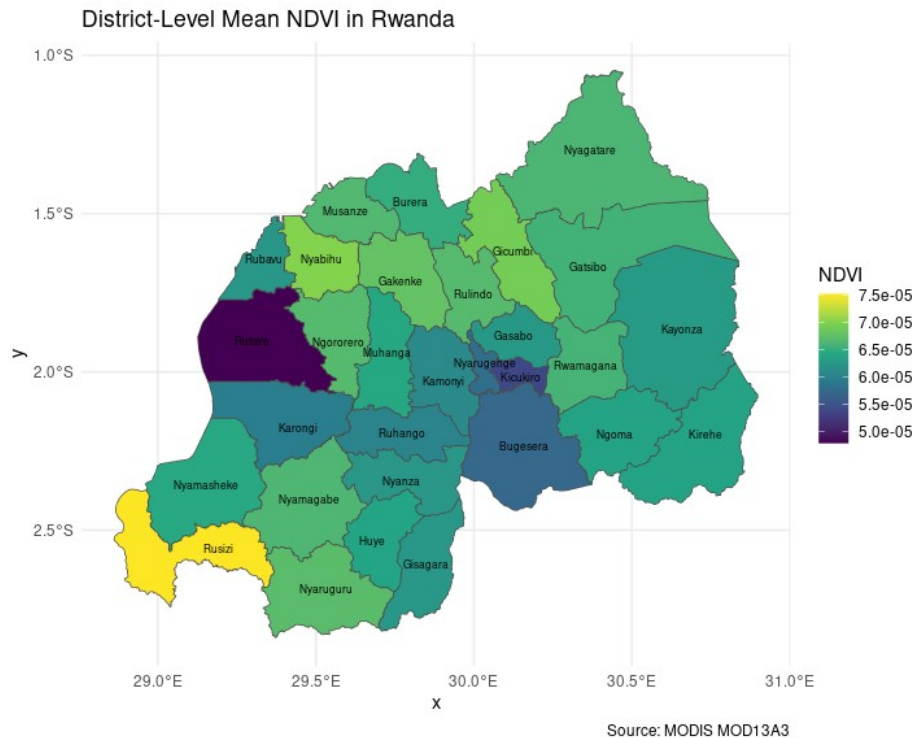
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
## 4.780e-05 6.238e-05 6.416e-05 6.359e-05 6.637e-05 7.519e-05
```

Map District-Level NDVI

```
centroids_ndvi <- st_point_on_surface(district_map)

ggplot(district_map) +
  geom_sf(aes(fill = ndvi_mean), color = "grey30", linewidth = 0.25) +
  geom_sf_text(data = centroids_ndvi, aes(label = NAME_2), size = 2.2)
+
  scale_fill_viridis_c(option = "viridis", name = "NDVI") +
  labs(
    title = "District-Level Mean NDVI in Rwanda",
```

```
caption = "Source: MODIS MOD13A3"
) +
theme_minimal()
```



Save Intermediate Datasets

```
write.csv(rainfall_district,
"~/Downloads/rainfall_district_final.csv", row.names = FALSE)
write.csv(ndvi_district,      "~/Downloads/ndvi_district_final.csv",
row.names = FALSE)
cat("Saved rainfall and NDVI district summaries.\n")

## Saved rainfall and NDVI district summaries.
```

STEP 15: Build the Final District-Level Modelling Dataset

This is the key dataset for all spatial models. It combines: - DHS-derived malnutrition rates and sociodemographic covariates - CHIRPS district-level mean rainfall - MODIS district-level mean NDVI

NOTE: We load from saved CSVs here so the model section can be run independently.

```

# Load saved datasets (adjust paths as needed)
NDVI_data      <- read_csv("~/Downloads/ndvi_district_final.csv")
rainfall_data  <- read_csv("~/Downloads/rainfall_district_final.csv")
RDHS_data      <- read_csv("~/Downloads/RDHS_final_clean.csv")

# --- Aggregate RDHS to district level ---
RDHS_sf <- st_as_sf(
  RDHS_data,
  coords = c("longitude", "latitude"),
  crs     = 4326
)
RDHS_sf <- st_transform(RDHS_sf, st_crs(rwanda_sf))

RDHS_district <- st_join(RDHS_sf, rwanda_sf[, c("NAME_2")])

rdhs_district_summary <- RDHS_district %>%
  st_drop_geometry() %>%
  group_by(NAME_2) %>%
  summarise(
    stunting_rate      = mean(stunted,      na.rm = TRUE) * 100,
    wasting_rate       = mean(wasted,       na.rm = TRUE) * 100,
    underweight_rate   = mean(underweight,  na.rm = TRUE) * 100,
    mean_wealth         = mean(wealth_idx,   na.rm = TRUE),
    mean_mat_edu        = mean(mat_edu,      na.rm = TRUE),
    mean_mat_bmi        = mean(mat_bmi,      na.rm = TRUE),
    mean_anc            = mean(anc_visits,   na.rm = TRUE),
    mean_altitude       = mean(altitude,     na.rm = TRUE),
    n_children          = n(),
    .groups             = "drop"
  )

# --- Join everything into final spatial dataset ---
model_data <- rwanda_sf %>%
  left_join(rdhs_district_summary,          by = "NAME_2") %>%
  left_join(rainfall_data[, c("NAME_2", "rainfall_mean")], by =
"NAME_2") %>%
  left_join(NDVI_data[, c("NAME_2", "ndvi_mean")],      by =
"NAME_2") %>%
  # Keep only districts with complete data for all model variables
  filter(
    !is.na(stunting_rate),
    !is.na(wasting_rate),
    !is.na(underweight_rate),

```

```

    !is.na(rainfall_mean),
    !is.na(ndvi_mean),
    !is.na(mean_altitude),
    !is.na(mean_wealth),
    !is.na(mean_mat_edu),
    !is.na(mean_mat_bmi),
    !is.na(mean_anc)
  )

cat("Districts in final model dataset:", nrow(model_data), "\n")

## Districts in final model dataset: 30

names(model_data)

## [1] "GID_2"          "GID_0"          "COUNTRY"
## [5] "NAME_1"         "NL_NAME_1"      "NAME_2"
## [9] "NL_NAME_2"      "TYPE_2"         "ENGTYPE_2"
## [13] "HASC_2"         "stunting_rate"  "wasting_rate"
## [17] "mean_wealth"    "mean_mat_edu"   "mean_mat_bmi"
## [21] "mean_altitude" "n_children"     "rainfall_mean"
## [25] "geometry"

```

Create Spatial Weights Matrix

A queen-contiguity spatial weights matrix is constructed. Districts sharing any boundary or corner are treated as neighbors. The matrix is row-standardized (style = "W") so each row sums to 1.

```

model_data <- st_make_valid(model_data)

nb <- poly2nb(model_data, queen = TRUE)
W <- nb2listw(nb, style = "W", zero.policy = TRUE)

summary(nb)

```



```
## Neighbour list object:
## Number of regions: 30
## Number of nonzero links: 138
## Percentage nonzero weights: 15.33333
## Average number of links: 4.6
## Link number distribution:
##
## 2 3 4 5 6 7
## 3 4 8 6 5 4
## 3 least connected regions:
## 17 19 25 with 2 links
## 4 most connected regions:
## 2 10 14 20 with 7 links
```

Global Moran's I : Spatial Autocorrelation Tests

Stunting

```
district_map_stunting <- district_map %>%
  st_make_valid() %>%
  filter(!is.na(stunting_rate))

nb_stunting <- poly2nb(district_map_stunting, queen = TRUE)
lw_stunting <- nb2listw(nb_stunting, style = "W", zero.policy = TRUE)

moran_stunting <- moran.test(
  district_map_stunting$stunting_rate,
  lw_stunting,
  zero.policy = TRUE
)
print(moran_stunting)

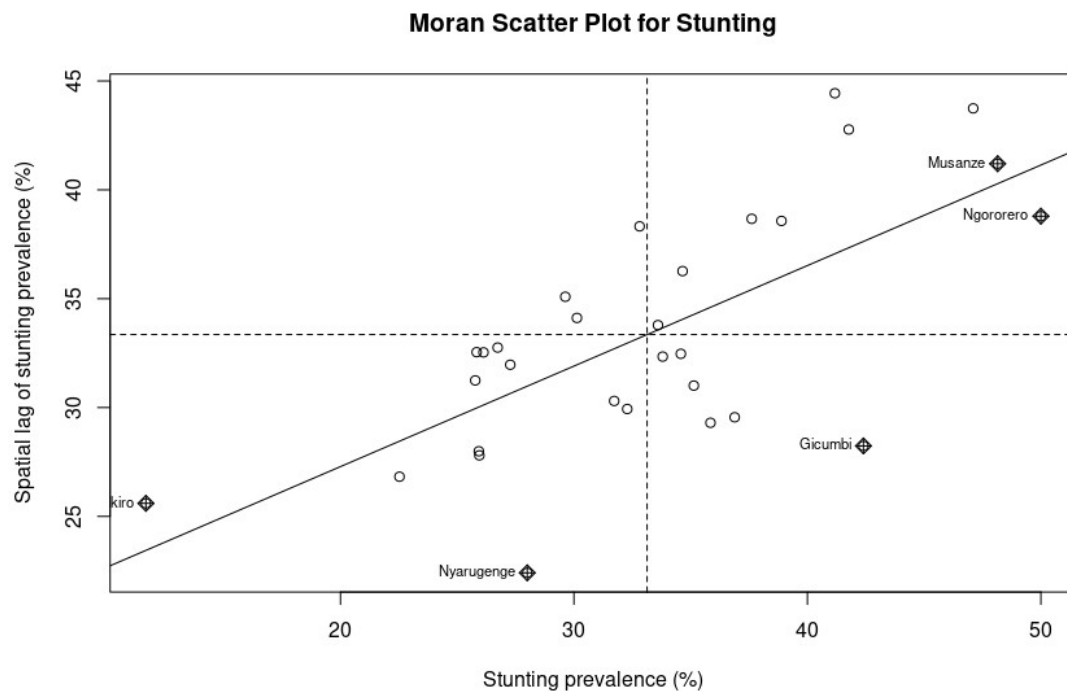
##
## Moran I test under randomisation
##
## data: district_map_stunting$stunting_rate
## weights: lw_stunting
##
## Moran I statistic standard deviate = 4.3632, p-value = 6.407e-06
## alternative hypothesis: greater
## sample estimates:
```

## Moran I statistic	Expectation	Variance
## 0.46172426	-0.03448276	0.01293323

```

moran.plot(
  district_map_stunting$stunting_rate,
  lw_stunting,
  zero.policy = TRUE,
  labels      = district_map_stunting$NAME_2,
  xlab        = "Stunting prevalence (%)",
  ylab        = "Spatial lag of stunting prevalence (%)",
  main        = "Moran Scatter Plot for Stunting"
)

```



Wasting

```

district_map_wasting <- district_map %>%
  st_make_valid() %>%
  filter(!is.na(wasting_rate))

nb_wasting <- poly2nb(district_map_wasting, queen = TRUE)
lw_wasting <- nb2listw(nb_wasting, style = "W", zero.policy = TRUE)

```

```

moran_wasting <- moran.test(
  district_map_wasting$wasting_rate,
  lw_wasting,
  zero.policy = TRUE
)
print(moran_wasting)

##
## Moran I test under randomisation
##
## data: district_map_wasting$wasting_rate
## weights: lw_wasting
##
## Moran I statistic standard deviate = 1.9194, p-value = 0.02746
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.16910934      -0.03448276      0.01125055

moran.plot(
  district_map_wasting$wasting_rate,
  lw_wasting,
  zero.policy = TRUE,
  labels      = district_map_wasting$NAME_2,
  xlab        = "Wasting prevalence (%)",
  ylab        = "Spatial lag of wasting prevalence (%)",
  main        = "Moran Scatter Plot for Wasting"
)

```



Underweight

```
district_map_underweight <- district_map %>%
  st_make_valid() %>%
  filter(!is.na(underweight_rate))

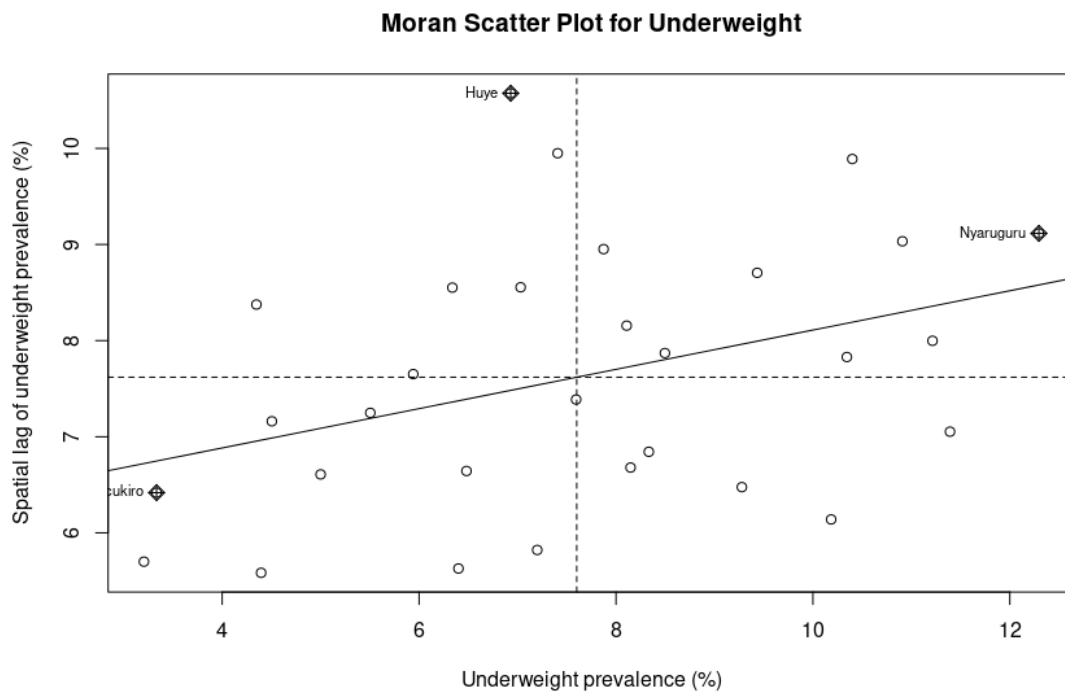
nb_underweight <- poly2nb(district_map_underweight, queen = TRUE)
lw_underweight <- nb2listw(nb_underweight, style = "W", zero.policy =
TRUE)

moran_underweight <- moran.test(
  district_map_underweight$underweight_rate,
  lw_underweight,
  zero.policy = TRUE
)
print(moran_underweight)

##
## Moran I test under randomisation
##
## data: district_map_underweight$underweight_rate
## weights: lw_underweight
##
## Moran I statistic standard deviate = 2.0532, p-value = 0.02003
## alternative hypothesis: greater
```

```
## sample estimates:
## Moran I statistic      Expectation      Variance
##           0.20445360      -0.03448276      0.01354283
```

```
moran.plot(
  district_map_underweight$underweight_rate,
  lw_underweight,
  zero.policy = TRUE,
  labels      = district_map_underweight$NAME_2,
  xlab        = "Underweight prevalence (%)",
  ylab        = "Spatial lag of underweight prevalence (%)",
  main        = "Moran Scatter Plot for Underweight"
)
```



LISA : Local Spatial Autocorrelation (Hotspot/Coldspot Analysis)

LISA classifies each district as:

- **High-High**: hotspot high malnutrition district surrounded by high-malnutrition neighbors
- **Low-Low**: coldspot low malnutrition district surrounded by low-malnutrition neighbors
- **High-Low**: spatial outlier high district surrounded by low neighbors
- **Low-High**: spatial outlier low district surrounded by high neighbors

- **Not significant:** no significant spatial clustering ($p \geq 0.05$)

LISA Helper Function

```
compute_lisa <- function(district_sf, rate_var, lw, alpha = 0.05) {
  lisa_result <- localmoran(district_sf[[rate_var]], lw, zero.policy =
TRUE)

  z_vals <- scale(district_sf[[rate_var]])[, 1]
  lag_z <- lag.listw(lw, z_vals, zero.policy = TRUE)
  p_vals <- lisa_result[, 5]

  district_sf$lisa_I <- lisa_result[, 1]
  district_sf$lisa_p <- p_vals
  district_sf$rate_z <- z_vals
  district_sf$lag_rate_z <- lag_z
  district_sf$lisa_cluster <- case_when(
    p_vals >= alpha ~ "Not significant",
    z_vals > 0 & lag_z > 0 ~ "High-High",
    z_vals < 0 & lag_z < 0 ~ "Low-Low",
    z_vals > 0 & lag_z < 0 ~ "High-Low",
    z_vals < 0 & lag_z > 0 ~ "Low-High",
    TRUE ~ "Not significant"
  )
  district_sf
}

lisa_colors <- c(
  "High-High" = "red",
  "Low-Low" = "blue",
  "High-Low" = "orange",
  "Low-High" = "lightblue",
  "Not significant" = "grey85"
)

plot_lisa_map <- function(data, title_text) {
  centroids <- st_point_on_surface(data)
  ggplot(data) +
    geom_sf(aes(fill = lisa_cluster), color = "grey30", linewidth =
0.25) +
    geom_sf_text(data = centroids, aes(label = NAME_2), size = 2) +
    scale_fill_manual(values = lisa_colors, name = "LISA Cluster") +
    labs(
      title = title_text,
```

```

    # subtitle = "Local Moran's I (p < 0.05)",
    caption = "Source: Rwanda DHS 2019/20"
  ) +
  theme_minimal()
}

```

LISA for Stunting

```

district_map_stunting <- compute_lisa(
  district_map_stunting, "stunting_rate", lw_stunting
)

cat("Stunting LISA cluster counts:\n")

## Stunting LISA cluster counts:

print(table(district_map_stunting$lisa_cluster))

##
##      High-High      Low-Low Not significant
##           6           5           19

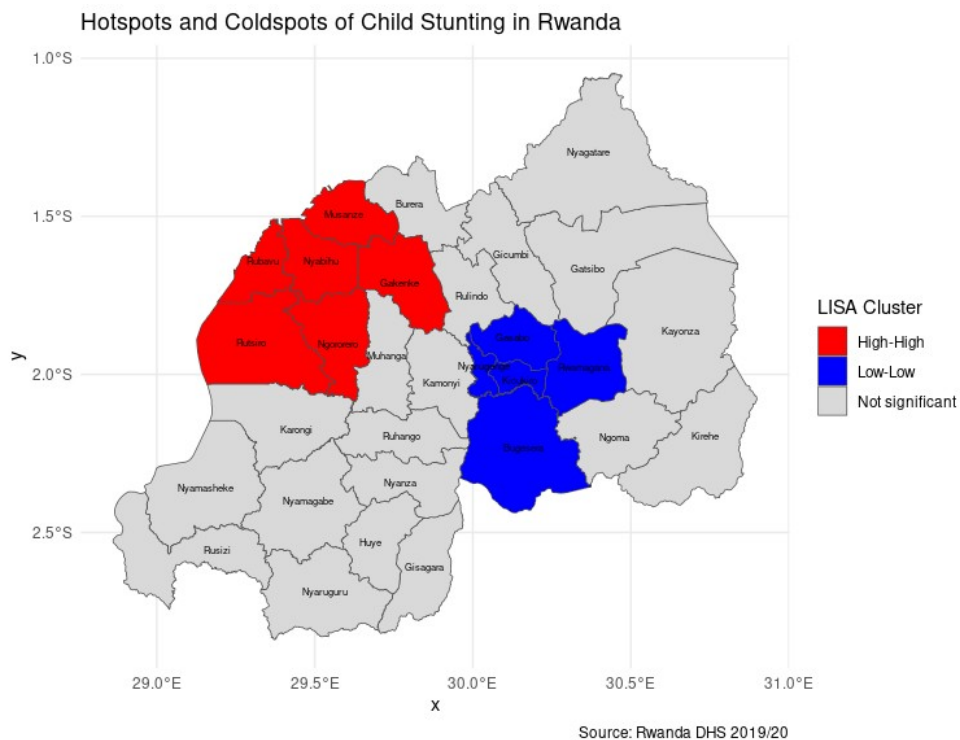
# Table of significant districts
district_map_stunting %>%
  st_drop_geometry() %>%
  filter(lisa_cluster != "Not significant") %>%
  select(NAME_2, stunting_rate, lisa_I, lisa_p, lisa_cluster) %>%
  arrange(lisa_cluster) %>%
  print()

##      NAME_2 stunting_rate  lisa_I  lisa_p lisa_cluster
## 1  Gakenke    37.61468 0.3772874 0.0384080920    High-High
## 2  Musanze    48.14815 1.8419667 0.0463188418    High-High
## 3  Ngororero   50.00000 1.4509008 0.0476390123    High-High

```

## 4	Nyabihu	47.10145	2.2531258	0.0006092282	High-High
## 5	Rubavu	41.17647	1.3823476	0.0396695069	High-High
## 6	Rutsiro	41.77215	1.2661333	0.0092860744	High-High
## 7	Bugesera	25.94937	0.5819616	0.0407991694	Low-Low
## 8	Rwamagana	25.92593	0.5632508	0.0480368154	Low-Low
## 9	Gasabo	22.52747	1.0173625	0.0439277430	Low-Low
## 10	Kicukiro	11.66667	2.4583785	0.0147889147	Low-Low
## 11	Nyarugenge	28.00000	0.8369744	0.0013077605	Low-Low

```
plot_lisa_map(district_map_stunting,
              "Hotspots and Coldspots of Child Stunting in Rwanda")
```



LISA for Wasting

```
district_map_wasting <- compute_lisa(
  district_map_wasting, "wasting_rate", lw_wasting
)

cat("Wasting LISA cluster counts:\n")

## Wasting LISA cluster counts:

print(table(district_map_wasting$lisa_cluster))
```

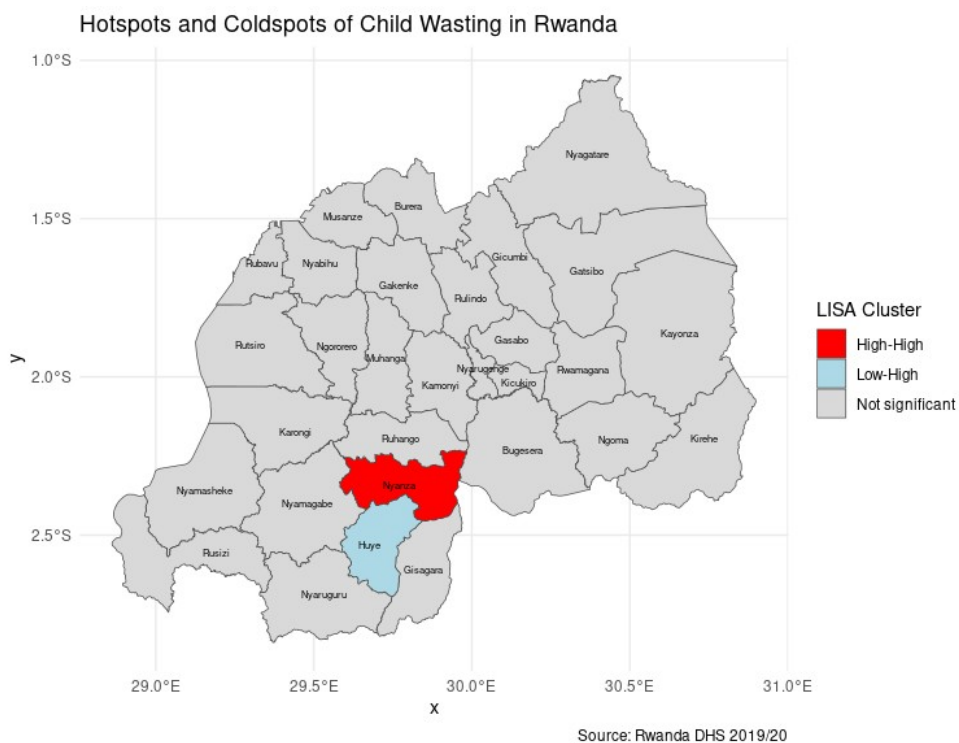


```
##
##           High-High           Low-High Not significant
##                1                 1                 28

district_map_wasting %>%
  st_drop_geometry() %>%
  filter(lisa_cluster != "Not significant") %>%
  select(NAME_2, wasting_rate, lisa_I, lisa_p, lisa_cluster) %>%
  arrange(lisa_cluster) %>%
  print()

##  NAME_2 wasting_rate    lisa_I    lisa_p lisa_cluster
## 1  Nyanza    1.574803  0.2931295 0.0383721052    High-High
## 2   Huye     0.000000 -1.6318737 0.0003734525    Low-High

plot_lisa_map(district_map_wasting,
              "Hotspots and Coldspots of Child Wasting in Rwanda")
```



LISA for Underweight

```
district_map_underweight <- compute_lisa(
  district_map_underweight, "underweight_rate", lw_underweight
)

cat("Underweight LISA cluster counts:\n")
```

```
## Underweight LISA cluster counts:
```

```
print(table(district_map_underweight$lisa_cluster))
```

```
##
```

```
##          Low-High          Low-Low Not significant
```

```
##              1              2              27
```

```
district_map_underweight %>%
```

```
  st_drop_geometry() %>%
```

```
  filter(lisa_cluster != "Not significant") %>%
```

```
  select(NAME_2, underweight_rate, lisa_I, lisa_p, lisa_cluster) %>%
```

```
  arrange(lisa_cluster) %>%
```

```
  print()
```

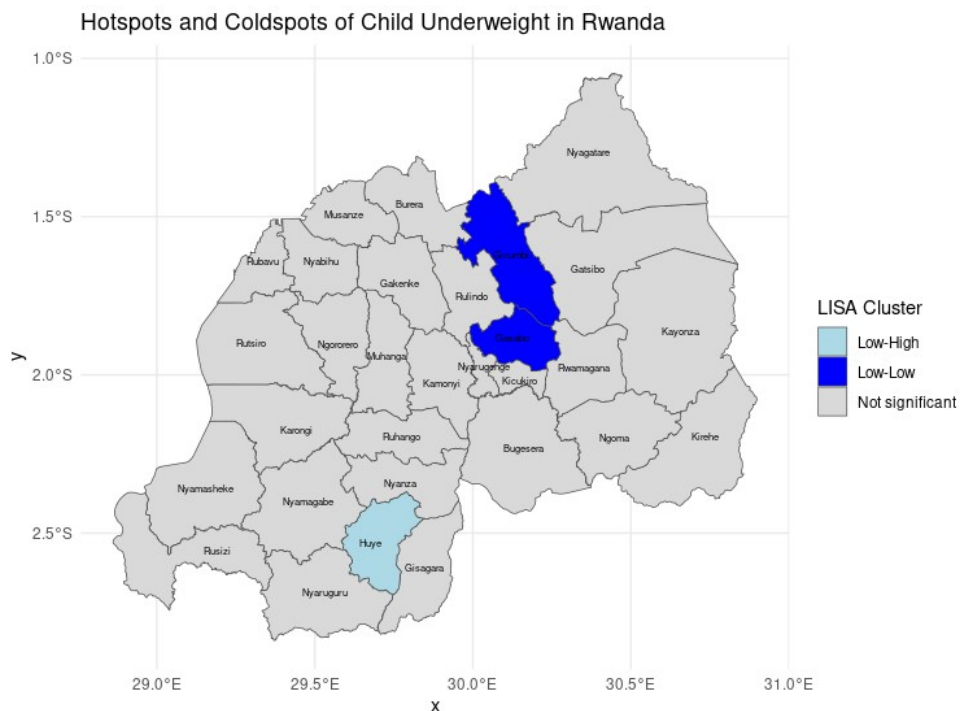
```
##    NAME_2 underweight_rate    lisa_I    lisa_p lisa_cluster
```

```
## 1    Huye          6.930693 -0.3362359 0.01158319    Low-High
```

```
## 2 Gicumbi          6.400000  0.3997657 0.02727649    Low-Low
```

```
## 3  Gasabo          4.395604  1.0911862 0.03230886    Low-Low
```

```
plot_lisa_map(district_map_underweight,  
              "Hotspots and Coldspots of Child Underweight in Rwanda")
```



OLS Baseline Models

An OLS regression is fitted first for each outcome as the baseline model. Residuals are tested for spatial autocorrelation (Moran's I on residuals). If significant spatial autocorrelation remains, spatial regression models are warranted.

OLS for Stunting

```
ols_stunting <- lm(
  stunting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data = model_data
)
summary(ols_stunting)

##
## Call:
## lm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
##     mean_altitude +
##     mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc, data =
##     model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.0190  -2.6610   0.5265   2.2833   8.5178
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.815e+01  4.189e+01   1.150  0.262671
## rainfall_mean -1.370e-02  1.039e-01  -0.132  0.896255
## ndvi_mean     2.534e+05  1.935e+05   1.310  0.203791
## mean_altitude 2.194e-02  5.626e-03   3.900  0.000769 ***
## mean_wealth    2.795e+00  4.695e+00   0.595  0.557753
## mean_mat_edu  -1.752e+01  9.247e+00  -1.895  0.071288 .
## mean_mat_bmi  -1.155e+00  1.988e+00  -0.581  0.567054
## mean_anc      -8.635e+00  4.409e+00  -1.958  0.062984 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.574 on 22 degrees of freedom
## Multiple R-squared:  0.7667, Adjusted R-squared:  0.6925
## F-statistic: 10.33 on 7 and 22 DF,  p-value: 1.064e-05

# Test OLS residuals for spatial autocorrelation
moran.test(residuals(ols_stunting), W, zero.policy = TRUE)
```

```
##
## Moran I test under randomisation
##
## data: residuals(ols_stunting)
## weights: W
##
## Moran I statistic standard deviate = 0.3056, p-value = 0.38
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.0004277871      -0.0344827586      0.0130497383
```

OLS — Wasting

```
ols_wasting <- lm(
  wasting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data = model_data
)
summary(ols_wasting)

##
## Call:
## lm(formula = wasting_rate ~ rainfall_mean + ndvi_mean +
##     mean_altitude +
##     mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc, data =
##     model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8053 -0.5826 -0.1857  0.3351  2.8728
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.079e+01  1.099e+01   1.892   0.0717 .
## rainfall_mean  2.324e-02  2.724e-02   0.853   0.4029
## ndvi_mean     -8.101e+04  5.076e+04  -1.596   0.1247
## mean_altitude -1.393e-03  1.476e-03  -0.944   0.3555
## mean_wealth    3.614e-02  1.232e+00   0.029   0.9769
## mean_mat_edu   1.807e+00  2.426e+00   0.745   0.4641
## mean_mat_bmi  -7.023e-01  5.215e-01  -1.347   0.1918
## mean_anc       2.987e-02  1.157e+00   0.026   0.9796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 1.2 on 22 degrees of freedom
## Multiple R-squared:  0.2886, Adjusted R-squared:  0.0623
## F-statistic: 1.275 on 7 and 22 DF,  p-value: 0.3074

moran.test(residuals(ols_wasting), W, zero.policy = TRUE)

##
## Moran I test under randomisation
##
## data: residuals(ols_wasting)
## weights: W
##
## Moran I statistic standard deviate = -0.42174, p-value = 0.6634
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      -0.08129908      -0.03448276      0.01232269
```

OLS — Underweight

```
ols_underweight <- lm(
  underweight_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
  mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data = model_data
)
summary(ols_underweight)

##
## Call:
## lm(formula = underweight_rate ~ rainfall_mean + ndvi_mean +
mean_altitude +
##      mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc, data =
model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.1747 -1.1238 -0.1589  1.0993  2.9809
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.957e+01  1.820e+01   1.075   0.2939
## rainfall_mean  7.460e-02  4.514e-02   1.653   0.1126
## ndvi_mean    -2.207e+05  8.410e+04  -2.624   0.0155 *
## mean_altitude -2.551e-03  2.445e-03  -1.043   0.3081
```

```
## mean_wealth    -2.803e+00  2.040e+00  -1.374    0.1833
## mean_mat_edu   -3.142e+00  4.019e+00  -0.782    0.4427
## mean_mat_bmi    5.019e-01  8.640e-01   0.581    0.5672
## mean_anc       -3.323e-01  1.916e+00  -0.173    0.8639
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.988 on 22 degrees of freedom
## Multiple R-squared:  0.5108, Adjusted R-squared:  0.3551
## F-statistic: 3.282 on 7 and 22 DF,  p-value: 0.01524

moran.test(residuals(ols_underweight), W, zero.policy = TRUE)

##
## Moran I test under randomisation
##
## data:  residuals(ols_underweight)
## weights: W
##
## Moran I statistic standard deviate = 1.6434, p-value = 0.05015
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.15447405      -0.03448276      0.01322056
```

Lagrange Multiplier (LM) Diagnostic Tests for Model Selection

LM tests guide the choice between Spatial Lag (LMlag) and Spatial Error (LMerr) models. The robust versions (RLMlag, RLMerr) are used when both standard tests are significant. The model corresponding to the more significant robust test is preferred.

LM Tests for Stunting

```
lm_tests_stunting <- lm.LMtests(
  ols_stunting, W,
  test      = c("LMlag", "LMerr", "RLMlag", "RLMerr"),
  zero.policy = TRUE
)
print(lm_tests_stunting)

##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
```

```

##
## data:
## model: lm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## RSlag = 1.1436, df = 1, p-value = 0.2849
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## RSerr = 1.1727e-05, df = 1, p-value = 0.9973
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## adjRSlag = 2.1097, df = 1, p-value = 0.1464
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +

```

```

mean_anc,
## data = model_data)
## test weights: listw
##
## adjRSerr = 0.96608, df = 1, p-value = 0.3257

```

LM Tests for Wasting

```

lm_tests_wasting <- lm.LMtests(
  ols_wasting, W,
  test      = c("LMlag", "LMerr", "RLMlag", "RLMerr"),
  zero.policy = TRUE
)
print(lm_tests_wasting)

##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = wasting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## RSlag = 0.0089485, df = 1, p-value = 0.9246
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = wasting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw

```



```
##
## RSerr = 0.42355, df = 1, p-value = 0.5152
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = wasting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
## mean_anc,
## data = model_data)
## test weights: listw
##
## adjRSlag = 3.6616, df = 1, p-value = 0.05568
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = wasting_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
## mean_anc,
## data = model_data)
## test weights: listw
##
## adjRSerr = 4.0762, df = 1, p-value = 0.04349
```

LM Tests for Underweight

```
lm_tests_underweight <- lm.LMtests(
  ols_underweight, W,
  test = c("LMlag", "LMerr", "RLMlag", "RLMerr"),
  zero.policy = TRUE
)
print(lm_tests_underweight)

##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
```

```

## dependence
##
## data:
## model: lm(formula = underweight_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## RSlag = 0.44809, df = 1, p-value = 0.5032
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = underweight_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## RSerr = 1.5291, df = 1, p-value = 0.2162
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = underweight_rate ~ rainfall_mean + ndvi_mean +
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
mean_anc,
## data = model_data)
## test weights: listw
##
## adjRSlag = 0.38504, df = 1, p-value = 0.5349
##
##
## Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial
## dependence
##
## data:
## model: lm(formula = underweight_rate ~ rainfall_mean + ndvi_mean +

```

```
## mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +  
mean_anc,  
## data = model_data)  
## test weights: listw  
##  
## adjRSerr = 1.4661, df = 1, p-value = 0.226
```

Spatial Lag Models

The Spatial Lag Model adds the spatially lagged dependent variable (Wy) as a predictor, capturing direct spillover of malnutrition between neighbouring districts. The spatial autoregressive parameter ρ measures the strength of this dependence.

```
model_data_scaled <- model_data %>%  
  mutate(  
    rainfall_mean = scale(rainfall_mean)[,1],  
    ndvi_mean     = scale(ndvi_mean)[,1],  
    mean_altitude = scale(mean_altitude)[,1],  
    mean_wealth   = scale(mean_wealth)[,1],  
    mean_mat_edu  = scale(mean_mat_edu)[,1],  
    mean_mat_bmi  = scale(mean_mat_bmi)[,1],  
    mean_anc      = scale(mean_anc)[,1]  
  )  
  
lag_stunting <- lagsarlm(  
  stunting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +  
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,  
  data          = model_data_scaled, # <-- use scaled data  
  listw         = W,  
  zero.policy   = TRUE  
)  
  
# Check correlations among predictors  
predictors <- model_data %>%  
  st_drop_geometry() %>%  
  select(rainfall_mean, ndvi_mean, mean_altitude,  
    mean_wealth, mean_mat_edu, mean_mat_bmi, mean_anc)  
  
cor_matrix <- cor(predictors, use = "complete.obs")  
print(round(cor_matrix, 2))
```

```
##                rainfall_mean ndvi_mean mean_altitude mean_wealth
mean_mat_edu
## rainfall_mean      1.00      0.45      0.59      -0.34
-0.29
## ndvi_mean          0.45      1.00      0.19      -0.29
-0.25
## mean_altitude      0.59      0.19      1.00      -0.40
-0.26
## mean_wealth        -0.34     -0.29     -0.40      1.00
0.85
## mean_mat_edu        -0.29     -0.25     -0.26      0.85
1.00
## mean_mat_bmi        -0.28     -0.31     -0.13      0.87
0.76
## mean_anc            0.29      0.28     -0.14      0.04
0.11
##                mean_mat_bmi mean_anc
## rainfall_mean      -0.28      0.29
## ndvi_mean          -0.31      0.28
## mean_altitude      -0.13     -0.14
## mean_wealth         0.87      0.04
## mean_mat_edu        0.76      0.11
## mean_mat_bmi        1.00     -0.17
## mean_anc           -0.17      1.00
```

Spatial Lag for stunting

```
lag_stunting <- lagsarlm(
  stunting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data      = model_data_scaled,
  listw     = W,
  zero.policy = TRUE
)
summary(lag_stunting)

##
## Call:lagsarlm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
##               mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
```

```
##      mean_anc, data = model_data_scaled, listw = W, zero.policy =
TRUE)
##
## Residuals:
##      Min      1Q   Median      3Q      Max
## -9.441361 -2.627397  0.096168  2.625741  8.165009
##
## Type: lag
## Coefficients: (asymptotic standard errors)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  27.02958    5.86545  4.6083  4.06e-06
## rainfall_mean -0.20563    1.10440 -0.1862  0.8522957
## ndvi_mean     1.29735    0.82812  1.5666  0.1172039
## mean_altitude  4.66027    1.23160  3.7839  0.0001544
## mean_wealth    1.16955    2.26286  0.5168  0.6052623
## mean_mat_edu   -2.71066    1.48952 -1.8198  0.0687856
## mean_mat_bmi   -1.17329    1.82801 -0.6418  0.5209790
## mean_anc       -1.98183    0.89046 -2.2256  0.0260390
##
## Rho: 0.183, LR test value: 1.1066, p-value: 0.29282
## Asymptotic standard error: 0.17287
##      z-value: 1.0586, p-value: 0.28979
## Wald statistic: 1.1206, p-value: 0.28979
##
## Log likelihood: -82.97687 for lag model
## ML residual variance (sigma squared): 14.675, (sigma: 3.8309)
## Number of observations: 30
## Number of parameters estimated: 10
## AIC: 185.95, (AIC for lm: 185.06)
## LM test for residual autocorrelation
## test value: 0.88269, p-value: 0.34747
```

Spatial Lag for Wasting

```
lag_wasting <- lagsarlm(
  wasting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data      = model_data_scaled,
  listw     = W,
  zero.policy = TRUE
```

```

)
summary(lag_wasting)

##
## Call:lagsarlm(formula = wasting_rate ~ rainfall_mean + ndvi_mean +
##      mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
##      mean_anc, data = model_data_scaled, listw = W, zero.policy =
TRUE)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -1.85279 -0.56766 -0.18493  0.34024  2.85843
##
## Type: lag
## Coefficients: (asymptotic standard errors)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.1210624   0.3416562   3.2813 0.001033
## rainfall_mean  0.2778296   0.2926782   0.9493 0.342485
## ndvi_mean     -0.4081647   0.2224291  -1.8350 0.066501
## mean_altitude -0.3229572   0.3117594  -1.0359 0.300240
## mean_wealth    0.0335321   0.6067543   0.0553 0.955928
## mean_mat_edu   0.3205434   0.3853749   0.8318 0.405539
## mean_mat_bmi  -0.7678814   0.4903350  -1.5660 0.117341
## mean_anc       0.0041632   0.2393722   0.0174 0.986124
##
## Rho: 0.031015, LR test value: 0.01184, p-value: 0.91335
## Asymptotic standard error: 0.24627
##      z-value: 0.12594, p-value: 0.89978
## Wald statistic: 0.015862, p-value: 0.89978
##
## Log likelihood: -43.37892 for lag model
## ML residual variance (sigma squared): 1.0553, (sigma: 1.0273)
## Number of observations: 30
## Number of parameters estimated: 10
## AIC: 106.76, (AIC for lm: 104.77)
## LM test for residual autocorrelation
## test value: 3.8387, p-value: 0.050082

```

Spatial Lag for Underweight

```

lag_underweight <- lagsarlm(
  underweight_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data          = model_data_scaled,
  listw         = W,
  zero.policy   = TRUE
)
summary(lag_underweight)

##
## Call:lagsarlm(formula = underweight_rate ~ rainfall_mean +
ndvi_mean +
##      mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
##      mean_anc, data = model_data_scaled, listw = W, zero.policy =
TRUE)
##
## Residuals:
##      Min        1Q    Median        3Q       Max
## -3.90195 -1.18836 -0.16754  1.15309  3.11946
##
## Type: lag
## Coefficients: (asymptotic standard errors)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   6.333460   1.605987  3.9437 8.025e-05
## rainfall_mean  0.791283   0.514796  1.5371 0.124274
## ndvi_mean     -1.102537   0.366863 -3.0053 0.002653
## mean_altitude -0.507500   0.508919 -0.9972 0.318662
## mean_wealth    -1.551236   0.992155 -1.5635 0.117935
## mean_mat_edu   -0.557596   0.631109 -0.8835 0.376956
## mean_mat_bmi    0.559379   0.802695  0.6969 0.485881
## mean_anc       -0.085951   0.392016 -0.2193 0.826452
##
## Rho: 0.16635, LR test value: 0.5143, p-value: 0.47328
## Asymptotic standard error: 0.2067
##      z-value: 0.80477, p-value: 0.42095
## Wald statistic: 0.64766, p-value: 0.42095
##
## Log likelihood: -58.27477 for lag model
## ML residual variance (sigma squared): 2.8313, (sigma: 1.6827)
## Number of observations: 30
## Number of parameters estimated: 10
## AIC: 136.55, (AIC for lm: 135.06)

```

```
## LM test for residual autocorrelation
## test value: 2.2437, p-value: 0.13416
```

Spatial Error Models

The Spatial Error Model captures spatial dependence through unobserved factors (e.g., unmeasured regional conditions) that cluster geographically. The spatial error parameter λ measures the strength of this hidden clustering.

Spatial Error for Stunting

```
error_stunting <- errorsarlm(
  stunting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data          = model_data_scaled,
  listw         = W,
  zero.policy   = TRUE
)
summary(error_stunting)

##
## Call:errorsarlm(formula = stunting_rate ~ rainfall_mean + ndvi_mean +
+
##   mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
##   mean_anc, data = model_data_scaled, listw = W, zero.policy =
TRUE)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.01572  -2.66232   0.52381   2.28008   8.52045
##
## Type: error
## Coefficients: (asymptotic standard errors)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  33.13382    0.71664 46.2351 < 2.2e-16
## rainfall_mean -0.16640    1.10660 -0.1504  0.88047
## ndvi_mean     1.29074    0.84478  1.5279  0.12654
## mean_altitude  5.31893    1.16990  4.5465 5.455e-06
## mean_wealth    1.59655    2.30614  0.6923  0.48875
## mean_mat_edu  -3.23707    1.46418 -2.2108  0.02705
## mean_mat_bmi  -1.26427    1.86887 -0.6765  0.49873
## mean_anc      -2.08043    0.90990 -2.2864  0.02223
```



```
##
## Lambda: 0.002013, LR test value: 2.5831e-05, p-value: 0.99594
## Asymptotic standard error: 0.26666
##      z-value: 0.007549, p-value: 0.99398
## Wald statistic: 5.6988e-05, p-value: 0.99398
##
## Log likelihood: -83.53016 for error model
## ML residual variance (sigma squared): 15.345, (sigma: 3.9173)
## Number of observations: 30
## Number of parameters estimated: 10
## AIC: 187.06, (AIC for lm: 185.06)
```

Spatial Error for Wasting

```
error_wasting <- errorsarlm(
  wasting_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data      = model_data_scaled,
  listw     = W,
  zero.policy = TRUE
)
summary(error_wasting)

##
## Call:errorsarlm(formula = wasting_rate ~ rainfall_mean + ndvi_mean
+
##      mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
##      mean_anc, data = model_data_scaled, listw = W, zero.policy =
TRUE)
##
## Residuals:
##      Min      1Q   Median      3Q      Max
## -1.60513 -0.59481 -0.14556  0.37031  2.82140
##
## Type: error
## Coefficients: (asymptotic standard errors)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   1.181242   0.120033  9.8410 < 2e-16
## rainfall_mean  0.413659   0.231936  1.7835  0.07450
## ndvi_mean     -0.502060   0.200830 -2.4999  0.01242
## mean_altitude -0.507643   0.238988 -2.1241  0.03366
## mean_wealth   -0.191714   0.542549 -0.3534  0.72382
## mean_mat_edu   0.560057   0.346552  1.6161  0.10608
## mean_mat_bmi  -0.739887   0.460865 -1.6054  0.10840
```

```
## mean_anc      0.047853  0.264862  0.1807  0.85662
##
## Lambda: -0.49466, LR test value: 1.2961, p-value: 0.25493
## Asymptotic standard error: 0.28019
##      z-value: -1.7654, p-value: 0.077493
## Wald statistic: 3.1167, p-value: 0.077493
##
## Log likelihood: -42.73681 for error model
## ML residual variance (sigma squared): 0.96126, (sigma: 0.98044)
## Number of observations: 30
## Number of parameters estimated: 10
## AIC: 105.47, (AIC for lm: 104.77)
```

Spatial Error for Underweight

```
error_underweight <- errorsarlm(
  underweight_rate ~ rainfall_mean + ndvi_mean + mean_altitude +
    mean_wealth + mean_mat_edu + mean_mat_bmi + mean_anc,
  data      = model_data_scaled,
  listw     = W,
  zero.policy = TRUE
)
summary(error_underweight)

##
## Call:errorsarlm(formula = underweight_rate ~ rainfall_mean +
ndvi_mean +
##      mean_altitude + mean_wealth + mean_mat_edu + mean_mat_bmi +
##      mean_anc, data = model_data_scaled, listw = W, zero.policy =
TRUE)
##
## Residuals:
##      Min      1Q  Median      3Q      Max
## -3.430164 -1.059854  0.066984  1.357736  3.143474
##
## Type: error
## Coefficients: (asymptotic standard errors)
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   7.61762    0.53829 14.1515 < 2.2e-16
## rainfall_mean  1.10961    0.58025  1.9123  0.055841
## ndvi_mean     -1.16809    0.36179 -3.2286  0.001244
```

```
## mean_altitude -0.39615    0.57579 -0.6880  0.491441
## mean_wealth   -1.54994    1.00429 -1.5433  0.122752
## mean_mat_edu  -0.67676    0.63294 -1.0692  0.284959
## mean_mat_bmi   0.60138    0.78029  0.7707  0.440877
## mean_anc      -0.21713    0.33009 -0.6578  0.510671
##
## Lambda: 0.45817, LR test value: 2.3812, p-value: 0.1228
## Asymptotic standard error: 0.19899
##      z-value: 2.3025, p-value: 0.021309
## Wald statistic: 5.3013, p-value: 0.021309
##
## Log likelihood: -57.34133 for error model
## ML residual variance (sigma squared): 2.5363, (sigma: 1.5926)
## Number of observations: 30
## Number of parameters estimated: 10
## AIC: 134.68, (AIC for lm: 135.06)
```

Model Comparison Using AIC

The model with the **lowest AIC** provides the best trade-off between goodness of fit and model complexity and is selected as the preferred model.

AIC Comparison for Stunting

```
aic_stunting <- AIC(ols_stunting, lag_stunting, error_stunting)
print(aic_stunting)

##              df      AIC
## ols_stunting   9 185.0603
## lag_stunting  10 185.9537
## error_stunting 10 187.0603

cat("Best stunting model:", rownames(aic_stunting)
[which.min(aic_stunting$AIC)], "\n")

## Best stunting model: ols_stunting
```

AIC Comparison for Wasting

```

aic_wasting <- AIC(ols_wasting, lag_wasting, error_wasting)
print(aic_wasting)

##              df      AIC
## ols_wasting   9 104.7697
## lag_wasting  10 106.7578
## error_wasting 10 105.4736

cat("Best wasting model:", rownames(aic_wasting)
[which.min(aic_wasting$AIC)], "\n")

## Best wasting model: ols_wasting

```

AIC Comparison for Underweight

```

aic_underweight <- AIC(ols_underweight, lag_underweight,
error_underweight)
print(aic_underweight)

##              df      AIC
## ols_underweight   9 135.0638
## lag_underweight  10 136.5495
## error_underweight 10 134.6827

cat("Best underweight model:", rownames(aic_underweight)
[which.min(aic_underweight$AIC)], "\n")

## Best underweight model: error_underweight

```

STEP 24: Residual Diagnostics of Selected Models

After selecting the best model per outcome, residuals are tested to confirm that spatial autocorrelation has been adequately accounted for. A well-specified spatial model should produce non-significant Moran's I on its residuals.

*# Replace these with the model selected by AIC for each outcome
 # Example: if Spatial Error was best for stunting and wasting,
 # but Spatial Lag was best for underweight:*

```

cat("=== Residual diagnostics for best stunting model ===\n")

## === Residual diagnostics for best stunting model ===

```

```

moran.test(residuals(error_stunting), W, zero.policy = TRUE)

##
## Moran I test under randomisation
##
## data: residuals(error_stunting)
## weights: W
##
## Moran I statistic standard deviate = 0.30202, p-value = 0.3813
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      1.897444e-05      -3.448276e-02      1.305010e-02

cat("=== Residual diagnostics for best wasting model ===\n")

## === Residual diagnostics for best wasting model ===

moran.test(residuals(error_wasting), W, zero.policy = TRUE)

##
## Moran I test under randomisation
##
## data: residuals(error_wasting)
## weights: W
##
## Moran I statistic standard deviate = 0.053361, p-value = 0.4787
## alternative hypothesis: greater
## sample estimates:
## Moran I statistic      Expectation      Variance
##      -0.02856750      -0.03448276      0.01228865

cat("=== Residual diagnostics for best underweight model ===\n")

## === Residual diagnostics for best underweight model ===

moran.test(residuals(error_underweight), W, zero.policy = TRUE)

##
## Moran I test under randomisation
##
## data: residuals(error_underweight)
## weights: W
##
## Moran I statistic standard deviate = 0.54338, p-value = 0.2934
## alternative hypothesis: greater

```

```
## sample estimates:
## Moran I statistic      Expectation      Variance
##      0.02841991      -0.03448276      0.01340096
```

STEP 25: Summary Table of All Model Results

```
# Function to extract key results from spatial models
extract_spatial_coefs <- function(model, outcome_name, model_type) {
  coefs <- summary(model)$Coef
  result <- data.frame(
    Outcome      = outcome_name,
    Model        = model_type,
    Variable      = rownames(coefs),
    Estimate      = round(coefs[, "Estimate"], 3),
    Std_Error     = round(coefs[, "Std. Error"], 3),
    p_value       = round(coefs[, "Pr(>|z|)"], 4),
    Significant    = ifelse(coefs[, "Pr(>|z|)"] < 0.05, "*", ""),
    row.names     = NULL
  )
  result
}
```

```
# Build summary for all selected models
# (Update model names based on AIC selection in Step 23)
results_stunting <- extract_spatial_coefs(error_stunting,
"Stunting",      "Spatial Error")
results_wasting <- extract_spatial_coefs(error_wasting,
"Wasting",       "Spatial Error")
results_underweight <- extract_spatial_coefs(error_underweight,
"Underweight",   "Spatial Error")
```

```
all_results <- bind_rows(results_stunting, results_wasting,
results_underweight)
```

```
print(all_results)
```

```
##      Outcome      Model      Variable Estimate Std_Error
p_value
## 1      Stunting Spatial Error (Intercept)   33.134      0.717
0.0000
## 2      Stunting Spatial Error rainfall_mean  -0.166      1.107
0.8805
## 3      Stunting Spatial Error ndvi_mean     1.291      0.845
```

0.1265					
## 4	Stunting	Spatial Error	mean_altitude	5.319	1.170
0.0000					
## 5	Stunting	Spatial Error	mean_wealth	1.597	2.306
0.4887					
## 6	Stunting	Spatial Error	mean_mat_edu	-3.237	1.464
0.0270					
## 7	Stunting	Spatial Error	mean_mat_bmi	-1.264	1.869
0.4987					
## 8	Stunting	Spatial Error	mean_anc	-2.080	0.910
0.0222					
## 9	Wasting	Spatial Error	(Intercept)	1.181	0.120
0.0000					
## 10	Wasting	Spatial Error	rainfall_mean	0.414	0.232
0.0745					
## 11	Wasting	Spatial Error	ndvi_mean	-0.502	0.201
0.0124					
## 12	Wasting	Spatial Error	mean_altitude	-0.508	0.239
0.0337					
## 13	Wasting	Spatial Error	mean_wealth	-0.192	0.543
0.7238					
## 14	Wasting	Spatial Error	mean_mat_edu	0.560	0.347
0.1061					
## 15	Wasting	Spatial Error	mean_mat_bmi	-0.740	0.461
0.1084					
## 16	Wasting	Spatial Error	mean_anc	0.048	0.265
0.8566					
## 17	Underweight	Spatial Error	(Intercept)	7.618	0.538
0.0000					
## 18	Underweight	Spatial Error	rainfall_mean	1.110	0.580
0.0558					
## 19	Underweight	Spatial Error	ndvi_mean	-1.168	0.362
0.0012					
## 20	Underweight	Spatial Error	mean_altitude	-0.396	0.576
0.4914					
## 21	Underweight	Spatial Error	mean_wealth	-1.550	1.004
0.1228					
## 22	Underweight	Spatial Error	mean_mat_edu	-0.677	0.633
0.2850					
## 23	Underweight	Spatial Error	mean_mat_bmi	0.601	0.780
0.4409					
## 24	Underweight	Spatial Error	mean_anc	-0.217	0.330
0.5107					

Significant

1 *

2

3

4 *

5

6 *

7

8 *

9 *

10

11 *

12 *

13

14

15

16

17 *

18

19 *

20

21

22

23

24